Software Architecture (WS 26/27)
Assignment 4

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05.11.2026

Task 1: AI-based Service

Task: Add a new service to your existing software architecture. This service should serve as a facade to interact with an LLM using the RWTH KI:connect API. The service does not need to be a domain-driven hexagon; nor does it need to be implemented using Spring or Java.

Task 2: Domain-driven Hexagon

In the lecture you learned about tactical DDD and the hexagonal architecture pattern to implement one service. Use the structure of the example project *Buckpal*[1] from the book *Get Your Hands Dirty on Clean Architecture* by T. Hombergs as starting point for organizing your code into an architecturally expressive structure.

Task: Implement one new standalone **core service** as a layered domain-driven hexagon and add it to your system. The service should provide at least two business functionalities (use cases with a rich domain model). Provide a brief documentation of your domain entities, value objects, and aggregates in a document including your rationale.

Implementation hints and requirements:

- The service should be implemented using the Spring Boot framework. We recommend to use Maven as build system.
- There should be at least one point of an ingoing, synchronous HTTP request via a driving adapter to the service. This should be invoked 1) via CURL/PostMan or other external tools, and 2) via one of the existing two services that you developed in week 2.
- There should be one point of an outgoing, synchronous HTTP request via a driven adapter to your new AI-based service.
- All services should be running in their own Docker containers in the same Docker environment.
- There is no need to add a database to the services yet. We will investigate this aspect later.
- During the entire course, user interfaces for the software components are optional.

[1] `https://gitlab.git.nrw/rwth-swa/teaching/buckpal`

Task 2: Project idea pitches

Task: Prepare a 5-minutes pitch of your project idea (domain, workflows, context map, and your existing services). The pitches will be given in the exercise session on **November 12**.

Requirements/Hints:

- The project idea pitch presentations are **graded** and **mandatory**. We expect each group member to be present for their group's pitch. The first slide of your presentation should have the group number, group name, and names of all group members.
- **Upload your slides (PDF only, 16:9 ratio, RWTH template) to the corresponding assignment in Moodle until November 11 (end of day, Aachen time).** Name them: **swa-26-idea-pitch-<moodle-group-id>-<groupname>.pdf**
- To ensure a smooth transition between groups, we will download all presentations to one of our computers and prepare them to be shown in the exercise session. The presentations will be given in ascending order according to the group number. There will be no form of (laser)pointer or remote control available.
- Please respect the time limits of 5 minutes per presentation.